

Lec19

Javascript in Browser

JavaScript makes websites **interactive**.

Without JS:

- Webpage only shows content (HTML)
- Styling through CSS

With JS:

- Button click
- Form validation
- Popup messages
- Dynamic updates

DOM (Document Object Model)

DOM means:

HTML page converted into JavaScript objects.

Browser reads HTML and creates a tree structure.

Example

<html>

<body>

<h1>Hello</h1>

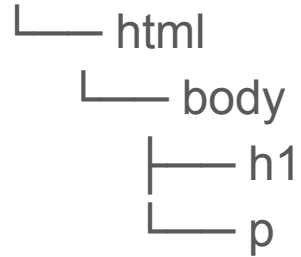
<p>Welcome</p>

</body>

</html>

Browser converts it into:

Document



This structure is called **DOM Tree**.

Using JS, we can:

- Access elements
- Change text
- Change styles
- Add/remove elements

Example:

```
document.body.style.background = "yellow";
```

Why DOM?

DOM allows JavaScript to:

- Read HTML
- Modify HTML
- Modify CSS
- React to user actions

Example:

```
document.getElementById("title").innerText = "New Heading";
```

BOM (Browser Object Model)

BOM allows JavaScript to interact with browser.

DOM → deals with webpage

BOM → deals with browser

Example

- Browser window size
- Browser history
- URL
- Alerts
- Timers

Important BOM objects:

- window
- navigator
- location
- history
- screen

DOM Element Selection

Before changing anything, first select element.

Example HTML:

```
<h1 id="heading">Hello</h1>
```

```
<p class="para">Paragraph 1</p>
```

```
<p class="para">Paragraph 2</p>
```

getElementById()

Select by ID.

```
let element = document.getElementById("heading");  
console.log(element);
```

getElementsByClassName()

Returns collection.

```
let items = document.getElementsByClassName("para");  
console.log(items);
```

getElementsByTagName()

```
let p = document.getElementsByTagName("p");
```

querySelector()

Select first matching element.

```
document.querySelector("#heading");
```

```
document.querySelector(".para");
```

```
document.querySelector("p");
```

querySelectorAll()

Select all matching elements.

```
let all = document.querySelectorAll(".para");
```

DOM Manipulation

After selecting element, modify it.

Change Content

HTML:

```
<h1 id="title">Hello</h1>
```

JS:

```
let title = document.getElementById("title");
```

```
title.innerText = "Welcome";
```

Difference:

innerText

Only visible text

innerHTML

Can include HTML tags

Change Style

```
title.style.color = "red";
```

```
title.style.backgroundColor = "yellow";
```

Change Attributes

HTML:

```

```

JS:

```
document.getElementById("image").src = "new.jpg";
```

Add Class

```
element.classList.add("active");
```

Remove:

```
element.classList.remove("active");
```

Toggle:

```
element.classList.toggle("active");
```

Creating Elements

Create new HTML elements using JS.

```
let newPara = document.createElement("p");  
newPara.innerText = "New Paragraph";
```

Add into body:

```
document.body.appendChild(newPara);
```

Removing Elements

```
element.remove();
```


Events in Js

Events = actions performed by user or browser.

Examples:

- click
- mouseover
- keypress
- submit
- load

Examples:

- Clicking button
- Typing in input
- Hovering mouse

Event Handling

HTML:

```
<button id="btn">Click</button>
```

Method 1: Inline

```
<button onclick="showMessage()">Click</button>
```

```
function showMessage() {  
    alert("Clicked");  
}
```

Not recommended.

Method 2: Using JS

```
let btn = document.getElementById("btn");
```

```
btn.onclick = function() {  
  alert("Clicked");  
};
```

Method 3: addEventListener() (Best)

```
btn.addEventListener("click", function() {  
  alert("Clicked");  
});
```

Best practice.

Common Events

Click Event

```
btn.addEventListener("click", function() {  
    console.log("Button clicked");  
});
```

Mouse Events

```
element.addEventListener("mouseover", function() {  
    console.log("Mouse entered");  
});  
element.addEventListener("mouseout", function() {  
    console.log("Mouse left");  
});
```

Keyboard Events

```
document.addEventListener("keydown", function() {  
  console.log("Key Pressed");  
});
```

Input Event

```
input.addEventListener("input", function() {  
  console.log("Typing...");  
});
```

Submit Event

```
form.addEventListener("submit", function() {  
  console.log("Form Submitted");  
});
```

Event Object

Event listener gets an object.

```
btn.addEventListener("click", function(event) {  
  console.log(event);  
});
```

Useful properties:

- target
- type

Example:

```
btn.addEventListener("click", function(event) {  
  console.log(event.target);  
});
```

Event Bubbling

When child clicked:

- Child event runs
- Parent event runs

Example:

```
<div>  
  <button>Click</button>  
</div>
```

Click button:

- Button event fires
- Div event fires

This is event bubbling.

Browser Interaction Methods

alert()

Show message.

```
alert("Welcome");
```

prompt()

Take input.

```
let name = prompt("Enter name");  
console.log(name);
```

confirm()

Returns true/false.

```
let result = confirm("Are you sure?");  
console.log(result);
```


Important BOM Objects

location

Current URL information.

```
console.log(location.href);
```

Change URL:

```
location.href = "https://google.com";
```

screen

Screen information.

```
console.log(screen.width);  
console.log(screen.height);
```

history

Browser history.

Back:

```
history.back();
```

Forward:

```
history.forward();
```

navigator

Browser information.

```
console.log(navigator.userAgent);
```

Timers in BOM

setTimeout()

Runs once after delay.

```
setTimeout(function() {  
    alert("Hello");  
}, 3000);
```

After 3 seconds.

setInterval()

Runs repeatedly.

```
setInterval(function() {  
    console.log("Running");  
}, 2000);
```

Every 2 seconds.

Difference between DOM and BOM

DOM	BOM
Document Object Model	Browser Object Model
Works with webpage	Works with browser
Manipulates HTML/CSS	Controls browser features

Task

Create DOM-based interactions using buttons and inputs